10/3/25 | Stoc Proc / HW4 | Walker, Nathan

1. (30 pts) In this problem you will derive the very important result that we have used many times about conditional Gaussian random variables.

Let $\mathbf{X}$ and $\mathbf{Y}$ each be random vectors, jointly distributed as

$$\begin{bmatrix}\mathbf{X}\\ \mathbf{Y}\end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\boldsymbol{\mu}_x\\ \boldsymbol{\mu}_y\end{bmatrix}, \Sigma\right)$$

where

$$\Sigma = \begin{bmatrix}\Sigma_{xx} & \Sigma_{xy}\\ \Sigma_{yx} & \Sigma_{yy}\end{bmatrix}$$

We make observation $\mathbf{Y} = \mathbf{y}$, and want to find the posterior density resulting from this. The conditional density is

$$f_{X|Y}(\mathbf{x}|\mathbf{y}) = \frac{f_{XY}(\mathbf{x},\mathbf{y})}{f_Y(\mathbf{y})} \tag{1}$$

In this problem you will derive $f_{X|Y}(\mathbf{x}|\mathbf{y})$, using the following steps.

(a) (5 pts) Write down $f_{XY}(\mathbf{x},\mathbf{y})$ in terms of $\boldsymbol{\mu}_x$, $\boldsymbol{\mu}_y$ and Σ. Write down $f_Y(\mathbf{y})$ in terms of $\boldsymbol{\mu}_y$ and Σ_{yy}.

(b) (10 pts) The matrix Σ can be factored to give a block diagonal form. That is, we can write

$$\begin{bmatrix}I & A\\ 0 & I\end{bmatrix}\Sigma\begin{bmatrix}I & 0\\ A^T & I\end{bmatrix} = \begin{bmatrix}B & 0\\ 0 & \Sigma_{yy}\end{bmatrix}$$

Determine A and B that make this true. (Think Gaussian elimination.)

(c) (5 pts) Based on this factorization, show that the quadratic piece of the exponent in $f_{XY}(\mathbf{x},\mathbf{y})$ can be written as

$$\begin{bmatrix}\mathbf{x}-\boldsymbol{\mu}_x\\ \mathbf{y}-\boldsymbol{\mu}_y\end{bmatrix}^T\Sigma^{-1}\begin{bmatrix}\mathbf{x}-\boldsymbol{\mu}_x\\ \mathbf{y}-\boldsymbol{\mu}_y\end{bmatrix}$$
$$= \begin{bmatrix}\mathbf{x}-\boldsymbol{\mu}_x\\ \mathbf{y}-\boldsymbol{\mu}_y\end{bmatrix}^T\begin{bmatrix}I & 0\\ -\Sigma_{yy}^{-1}\Sigma_{xy}^T & I\end{bmatrix}\begin{bmatrix}(\Sigma_{xx}-\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx})^{-1} & 0\\ 0 & \Sigma_{yy}^{-1}\end{bmatrix}\begin{bmatrix}I & -\Sigma_{xy}\Sigma_{yy}^{-1}\\ 0 & I\end{bmatrix}\begin{bmatrix}\mathbf{x}-\boldsymbol{\mu}_x\\ \mathbf{y}-\boldsymbol{\mu}_y\end{bmatrix}$$

(d) (5 pts) Further show that this quadratic piece of the exponent can be written as

$$= (\mathbf{x}-\bar{\mathbf{x}})^T(\Sigma_{xx}-\Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx})^{-1}(\mathbf{x}-\bar{\mathbf{x}}) + (\mathbf{y}-\boldsymbol{\mu}_y)^T\Sigma_{yy}^{-1}(\mathbf{y}-\boldsymbol{\mu}_y) + \text{stuff}$$

where "stuff" does not depend on $\mathbf{x}$. In this expression, determine the value of $\bar{\mathbf{x}}$.

(e) (5 pts) Using this, write down the conditional density $f_{X|Y}(\mathbf{x}|\mathbf{y})$. Identify the mean and the covariance of this conditional distribution.

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a)

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{\det\Sigma}}\exp\left(-\frac{1}{2}\begin{bmatrix}x-\mu_x\\ y-\mu_y\end{bmatrix}^T\Sigma^{-1}\begin{bmatrix}x-\mu_x\\ y-\mu_y\end{bmatrix}\right)$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_{yy}}}\exp\left[\frac{-(y-\mu_y)^2}{2\sigma_{yy}}\right]$$

b)

$$\begin{bmatrix}I & A\\ 0 & I\end{bmatrix}\begin{bmatrix}\sigma_{xx} & \sigma_{xy}\\ \sigma_{yx} & \sigma_{yy}\end{bmatrix}\begin{bmatrix}I & 0\\ A & I\end{bmatrix} = \begin{bmatrix}B & 0\\ 0 & \sigma_{yy}\end{bmatrix}$$

$$\begin{bmatrix}\sigma_{xx}+\sigma_{yx}A & \sigma_{xy}+\sigma_{yy}A\\ \sigma_{yx} & \sigma_{yy}\end{bmatrix}\begin{bmatrix}I & 0\\ A & I\end{bmatrix} = \begin{bmatrix}\sigma_{xx}+\sigma_{yx}A+(\sigma_{xy}+\sigma_{yy}A)A & \sigma_{xy}+\sigma_{yy}A\\ \sigma_{yx}+A\sigma_{yy} & \sigma_{yy}\end{bmatrix}$$

$$A = \frac{-\sigma_{xy}}{\sigma_{yy}} \qquad B = \sigma_{xx} - \frac{\sigma_{yx}^2}{\sigma_{yy}}$$

c)

2. (5 pts) Box Muller: Let $X_1 \sim \mathcal{U}(0,1)$ and $X_2 \sim \mathcal{U}(0,1)$ (independent). Let

$$Y_1 = \sqrt{-2\ln X_1}\cos(2\pi X_2) \qquad Y_2 = \sqrt{-2\ln X_1}\sin(2\pi X_2)$$

Show that

$$Y_1 \sim \mathcal{N}(0,1) \qquad Y_2 \sim \mathcal{N}(0,1).$$

$R = \sqrt{-2\ln x_1}$ $\quad \Theta = 2\pi X_2$ $\quad X_1, X_2 = 1$ for $0 < x_1, x_2 < 1$

$Y_1 = R\cos\Theta$ $\quad Y_2 = R\sin\Theta$

$x_1 = e^{-\frac{r^2}{2}}$ $\quad x_2 = \frac{\theta}{2\pi}$ $\quad \frac{\partial x_1}{\partial r} = -re^{-r^2/2}$ $\quad x_2 = \frac{1}{2\pi}$

$J = \left|\det \frac{\partial(u_1,u_2)}{\partial(r,\theta)}\right| = \frac{re^{-r^2/2}}{2\pi}$

$F_{R,\Theta}(r,\theta) = \frac{re^{-r^2/2}}{2\pi}$

$F_{Y_1,Y_2}(y_1,y_2) = \frac{1}{r}F_{r,\theta}(r,\theta) = \frac{1}{2\pi}e^{\frac{-(y_1^2+y_2^2)}{2}} = \frac{1}{2\pi}\left(e^{\frac{-y_1^2}{2}}\cdot e^{\frac{-y_2^2}{2}}\right)$

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3. (5 pts) If X and Y are independent and $Y \sim \mathcal{U}(0,1)$, show that $Z = X + Y$ has the density

$$f_Z(z) = F_x(z) - F_x(z-1).$$

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addition is conv of pdfs

$f(x) = \int_{-\infty}^{\infty} f_x(z-y)f_y(y)\,dy$ $\quad f(y)$ exists from 0 to 1

$P(0 < z \le 1) = \int_0^1 f_x(z-y)\,dy$

$= F(z) - F(z-1)$

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4. (5 pts) Let $X \sim \mathcal{N}(0,\sigma^2)$ and $Y \sim \mathcal{N}(0,\sigma^2)$ be independent. Let $Z = X^2 + Y^2$. Show that

$$f_Z(z) = \frac{1}{2\sigma^2}e^{-z/2\sigma^2}u(z)$$

If $\sigma^2 = 1$, the random variable Z is said to be chi-squared (χ^2) distributed with two degrees of freedom.

$f_{x,y}(x,y) = \frac{1}{2\pi\sigma^2}e^{\frac{-x^2-y^2}{2\sigma^2}}$ $\quad r^2 = x^2 - y^2$

$J = r$

$f_{R,\theta}(r,\theta) = \frac{r}{2\pi\sigma^2}e^{\frac{-r^2}{2\sigma^2}}$

$f_R(r) = \int_0^{2\pi}\frac{r}{2\pi\sigma^2}e^{\frac{-r^2}{2\sigma^2}}d\theta = \frac{r}{\sigma^2}e^{\frac{-r^2}{2\sigma^2}}$

$z = x^2+y^2 = R^2$ $\quad z = r^2$ $\quad r = \sqrt{z}$ $\quad \frac{dr}{dz} = \frac{1}{2\sqrt{z}}$

$f_z(z) = f_R(\sqrt{z})\frac{dr}{dz} = \frac{\sqrt{z}}{\sigma^2}e^{\frac{-z}{2\sigma^2}}\cdot\frac{1}{2\sqrt{z}} = \frac{1}{2\sigma^2}e^{\frac{-z}{2\sigma^2}}u(z)$

5. (5 pts) If $X \in \mathcal{U}(0,1)$ and $Y = -(\ln(X))/\lambda$, show that Y is exponentially distributed.

$g(x) = -\frac{\ln x}{\lambda} = Y \Rightarrow X = e^{-\lambda y}$ $\quad \frac{dx}{dy} = -\lambda e^{-\lambda y}$

$F_Y(y) = P(Y \le y) = p(-\frac{\ln x}{\lambda} \le y) = p(X \ge e^{-\lambda y})$

$F_Y(y) = 1 - e^{-\lambda y}$ $\quad y \ge 0$ $\quad x = e^{-\lambda y}$

$\left|\frac{dx}{dy}\right| = \lambda e^{-\lambda y}$

$f_Y(y) = f_x(x(y)) \left|\frac{dx}{dy}\right| = 1 \cdot \lambda e^{-\lambda y} = \lambda e^{-\lambda y}$

6. (10 pts) Let $g(x)$ be a monotone increasing function and let $Y = g(X)$. Show that

$$F_{XY}(x,y) = \begin{cases} F_X(x) & \text{if } y > g(x) \\ F_Y(y) & \text{if } y < g(x). \end{cases}$$

$F_{X,Y}(x,y) = P(X \le x, Y \le y) = P(X \le x, g(X) \le y)$

$= p(x \le x, X \le g^{-1}(y))$

$= p(X \le \min\{x, g^{-1}(y)\})$

$= F_X(\min(x, g^{-1}(y)))$

if $y > g(x)$ then $g^{-1}(y) > x$

min is x

if $y < g(x)$ then $g^{-1}(y) < x$

$\therefore$ the statement is true

7. (10 pts) Let $Z = aX + bY$. Show that

$$f_Z(z) = \frac{1}{|a|} \int_{-\infty}^{\infty} f_{XY}(\frac{z - by}{a}, y)\, dy.$$

Hint: Let $W = Y$ be an auxiliary variable.

$x = \frac{z - by}{a}$ $\quad W = Y$ $\quad J = \left|\begin{bmatrix} 1/a & -b/a \\ 0 & 1 \end{bmatrix}\right| = \frac{1}{|a|}$

$F_Z(z) = P(Z < z) = p(aX + bY < z)$

$f_{z,w}(z,w)\,(x(z,w), Y(z,w)) \cdot |J| = f_{x,w}(\frac{z - bw}{a}, w)\, \frac{1}{|a|}$

$f_Z(z) = \int_{-\infty}^{\infty} f_{z,w}(z,w)\, dw = \frac{1}{|a|} \int_{-\infty}^{\infty} f_{x,y}(\frac{z - by}{a}, y)\, dy$

8. (10 pts) Let X and Y be independent with

$$f_X(x) = \alpha e^{-\alpha x} u(x) \qquad f_Y(y) = \beta e^{-\beta y} u(y).$$

Determine the density of:

(a) (5 pts) $Z = X/Y$.

(b) (5 pts) $Z = \max(X, Y)$.

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a) $X = ZY$

$$f_Z(z) = \int_0^\infty y f_X(zy) f_Y(y)\, dy$$

$$= \int_0^\infty y \alpha e^{-zy\alpha} \beta e^{-y\beta}\, dy$$

$$= \alpha\beta \int_0^\infty y e^{-y(z\alpha+\beta)}\, dy \qquad k = z\alpha + \beta$$

$$= \alpha\beta \frac{1}{k^2}$$

$$f_Z(z) = \frac{\alpha\beta}{(z\alpha+\beta)^2} \qquad z \geq 0$$

b)

$$F_Z(z) = P(Z \leq z) = P(\max(X,Y) \leq z)$$

$$= P(X \leq z, Y \leq z)$$

$$= P(X \leq z) P(Y \leq z)$$

$$= F_X(z) F_Y(z)$$

$$F_X(z) = 1 - e^{-\alpha z} \qquad F_Y(z) = 1 - e^{-\beta z}$$

$$\frac{d}{dz} F_Z(z) = 1 - e^{-\alpha z} - e^{-\beta z} + e^{-z(\alpha+\beta)}$$

$$f_Z(z) = \alpha e^{-\alpha z} + \beta e^{-\beta z} - (\alpha+\beta) e^{-z(\alpha+\beta)} \qquad z \geq 0$$

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9. (5 pts) Show that if X, Y are i.i.d. $\mathcal{N}(0, \sigma^2)$, then $Z = X/Y$ has a Cauchy distribution,

$$f_Z(z) = \frac{1/\pi}{z^2+1}.$$

$$f_Z(z) = \int_{-\infty}^{\infty} |y| f_X(zy) f_Y(y)\, dy$$

$$= \int_{-\infty}^{\infty} \frac{|y|}{2\pi\sigma^2} \exp\left[-\frac{(zy)^2+y^2}{2\sigma^2}\right] dy$$

$$= \int_{-\infty}^{\infty} \frac{|y|}{2\pi\sigma^2} \exp\left[\frac{-y^2(z^2+1)}{2\sigma^2}\right] dy \qquad \text{even}$$

$$= \frac{1}{\pi\sigma^2} \int_0^\infty y \exp\left[\frac{-y^2(z^2+1)}{2\sigma^2}\right] dy$$

$$= \frac{1}{\pi(z^2+1)} \int_0^\infty e^{-u}\, du$$

$$= \frac{y\pi}{z^2+1}$$

$$u = \frac{-y^2(z^2+1)}{2\sigma^2}$$

$$du = \frac{-y(z^2+1)\, dy}{\sigma^2}$$

$$\frac{\sigma^2}{z^2+1} du = y\, dy$$

10. (5 pts) G&S Exercise 4.4.3: Let X have the uniform distribution on $[0, 1]$. For what function g does $Y = g(X)$ have the exponential distribution with parameter 1?

$$Y = -\ln X$$

$$F_Y(y) = P(Y \le y) = P(-\ln X \le y)$$
$$= P(X \ge e^{-y})$$
$$= 1 - F_X(e^{-y})$$
$$\frac{d}{dy} F_Y(y) = 1 - e^{-y}$$
$$f_Y(y) = e^{-y}$$

11. (5 pts) G&S Exercise 4.7.3: Let X be uniformly distributed on $[0, \pi/2]$. Find the density function of $Y = \sin X$.

$$F_Y(y) = P(Y \le y) = P(\sin X \le y)$$
$$= P(X \le \sin^{-1} y)$$
$$= F_X(\sin^{-1} y)$$
$$\frac{d}{dy}\left(F_Y(y) = \sin^{-1} y \cdot \frac{2}{\pi}\right)$$
$$f_Y(y) = \frac{2}{\pi\sqrt{1-y^2}}$$

12. (10 pts) G&S Exercise 4.7.4: Find the density function of $Y = \sin^{-1} X$ when:

(a) (5 pts) X is uniformly distributed on $[0, 1]$.

(b) (5 pts) X is uniformly distributed on $[-1, 1]$.

$$F_Y(y) = P(Y \le y) = P(\sin^{-1} X \le y)$$
$$= P(X \le \sin y)$$
$$= F_X(\sin y)$$
$$\frac{d}{dy} F_Y(y) = \sin y \cdot \frac{1}{b-a}$$
$$f_Y = \cos y \frac{1}{b-a}$$

$X = U(0,1)$ $\quad f_Y = \cos Y \quad 0 \le Y \le \pi/2$

$X = U(-1,1)$ $\quad f_Y = \frac{1}{2}\cos y \quad -\frac{\pi}{2} \le Y \le \pi/2$

13. (5pts) G&S Exercise 4.8.1: Let X and Y be independent variables having the exponential distribution with parameters λ and μ, respectively. Find the density function of $Z = X + Y$.

Hint: Fourier transform

$f_X(x) = \lambda e^{-\lambda x}$ $\quad f_Y(y) = \mu e^{-\mu y}$

$\phi_X(u) = \int_0^\infty \lambda e^{-\lambda x} e^{i2\pi u x}\, dx$

$= \lambda \int_0^\infty e^{(i2\pi u - \lambda)x}\, dx$

$= \frac{\lambda}{\lambda - i2\pi u} e^{(i2\pi u + \lambda)-x}\Big|_x^\infty = \frac{\lambda}{\lambda - i2\pi u}$ $\quad \phi_Y(u) = \frac{\mu}{\mu - i2\pi u}$

$\phi_Z(u) = \frac{\lambda}{\lambda - i2\pi u} \cdot \frac{\mu}{\mu - i2\pi u} = \frac{\lambda\mu}{\mu - \lambda}\left(\frac{1}{\lambda - i2\pi u} - \frac{1}{\mu - i2\pi u}\right)$

using table

$f_Z(z) = \frac{\lambda\mu}{\mu - \lambda}\left(e^{-\lambda z} - e^{-\mu z}\right) u(z)$

14. (5 pts) G&S Exercise 4.8.2: Let X and Y be independent variables with the Cauchy distribution,

$$f_X(x) = \frac{1}{\pi(x^2+1)} \qquad f_Y(y) = \frac{1}{\pi(y^2+1)}.$$

Find the density function of $Z = \alpha X + \beta Y$ where $\alpha\beta \neq 0$.

Hint: Fourier transform.

$\phi_X(u) = e^{-|u|}$ $\quad \phi_Y(u) = e^{-|u|}$

$\phi_Z(u) = \alpha\beta e^{-2|u|}$

$f_Z(z) = \frac{2\alpha\beta}{\pi(4 + x^2)}$

15. (5 pts) Let

$$g(x) = \frac{1}{1+e^{-2x/\sigma^2}}$$

and suppose that $X \sim \mathcal{N}(1, \sigma^2)$. Show that the pdf of $Y = g(X)$ is

$$f_Y(y) = \frac{\sigma}{2\sqrt{2\pi}y(1-y)} \exp[-\frac{1}{2\sigma^2}(\frac{\sigma^2}{2}\ln\frac{y}{1-y} - 1)^2]$$

Make plots of $f_Y(y)$ for a few values of σ^2.

$$y = \frac{1}{1+e^{-2x/\sigma^2}}$$

$$1+e^{-2x/\sigma^2} = 1/y$$

$$e^{-2x/\sigma^2} = 1/y - 1$$

$$-2x/\sigma^2 = \ln 1/y - 1$$

$$x = -\frac{\sigma^2}{2}\ln 1/y - 1 \qquad \left|\frac{dx}{dy}\right| = -\frac{\sigma^2}{2}\frac{-1}{y} \cdot \frac{1}{y-1} = \left|\frac{-\sigma^2}{2y(1-y)}\right| = \frac{\sigma^2}{2y(1-y)}$$

$$f_x(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-(y-1)^2/2\sigma^2\right]$$

$$f_x(g^{-1}(y)) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\left(\frac{\sigma^2}{2}\ln(\tfrac{y}{1-y}) - 1\right)^2/2\sigma^2\right]$$

$$f_x(g^{-1}(y)) \cdot \left|\frac{dx}{dy}\right| = \frac{\sigma}{\sqrt{2\pi}y(1-y)} \exp\left[-\left(\frac{\sigma^2}{2}\ln(\tfrac{y}{1-y}) - 1\right)^2 \cdot \frac{1}{2\sigma^2}\right]$$

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clear all; clc; close all; format compact;

y = 0:.01:1;
sigma = [1, 1.5, 2, 2.5];

figure(1);
hold on;
for i = 1:4
    S = sigma(i);
    f = S./(2*sqrt(2*pi).*y.*(1.-y));
    temp = exp((S^2 * log((1./y) - 1) - 1).^2 ...
        / (-2*S^2));
    f = f .* temp;
    % figure(i)
    plot(abs(f));
end
legend(arrayfun(@(s) sprintf('\\sigma = %.1f', s), ...
    sigma, 'UniformOutput', false));
```

